$$I = \int_1^2 \left[\frac{-1}{(x+1)} + \frac{2}{x+2}\right] dx \qquad \text{(Using partial fractions)}$$

$$= \left[-\log(x+1) + 2\log(x+2)\right]_1^2$$

$$= \log\left[\frac{(x+2)^2}{x+1}\right]_1^2$$

$$= \log\frac{16}{3} - \log\frac{9}{2}$$

$$= \log\frac{32}{27}.$$

Example 9.11

Evaluate : $\int_0^{\frac{\pi}{2}} \frac{\cos\theta}{(1+\sin\theta)(2+\sin\theta)} d\theta$.

Solution

Let $I = \int_0^{\frac{\pi}{2}} \frac{\cos\theta}{(1+\sin\theta)(2+\sin\theta)} d\theta$. Put $u = 1+\sin\theta$. Then, $du = \cos\theta d\theta$.

When $\theta = 0$, $u = 1$. When $\theta = \frac{\pi}{2}$, $u = 2$.

$$\therefore I = \int_1^2 \frac{du}{u(1+u)} = \int_1^2 \frac{(1+u)-u}{u(1+u)} du = \int_1^2 \left(\frac{1}{u} - \frac{1}{1+u}\right) du = \left[\log u - \log(1+u)\right]_1^2$$

$$= (\log 2 - \log 3) - (\log 1 - \log 2) = 2\log 2 - \log 3 = \log\frac{4}{3}.$$

Example 9.12

Evaluate : $\int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx$.

Solution

Let $I = \int_0^{\frac{1}{\sqrt{2}}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx$.

Put $u = \sin^{-1} x$. Then, $x = \sin u$ and so, $du = \frac{1}{\sqrt{1-x^2}} dx$.

When $x = 0$, $u = 0$. When $x = \frac{1}{\sqrt{2}}$, $u = \frac{\pi}{4}$.

$$\therefore I = \int_0^{\frac{\pi}{4}} \frac{u}{\cos^2 u} du = \int_0^{\frac{\pi}{4}} u\sec^2 u \, du = [u\tan u]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan u \, du = \left[u\tan u\right]_0^{\frac{\pi}{4}} + \left[\log\cos u\right]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} + \log\frac{1}{\sqrt{2}} = \frac{\pi}{4} - \frac{1}{2}\log 2.$$

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